Matched premium AC validation, version 2

Setup and identities

All 18 original selected UTC hours, three original state IDs per hour and unity/0.98 lagging/0.98 leading assumptions are retained. Four fixed policies are evaluated: ordinary rolling M2, raw U, raw C and retrospective EC. Two separately labeled mappings produce 648 cases each, 1,296 new campaign solves; the independent 27-case preflight adds 27 historical baseline solves. No additional cases are counted unless actually evaluated. Exact cases and configurations are in MATCHED_AC_CASES.csv and AC_CONFIGURATION_V2.json.

The historical two-map configuration preserves the original reactive structural matrix. Common v2 sets C_Q=C_P_final by an explicit participant-level PF modeling decision. It is not a recovery of undocumented physical device locations. Every other scientific input, including fixed PV reference dispatch and original rating conventions, remains unchanged.

The baseline sample was predeclared before solving. All 27 historical AC records and nine independent DC cases pass. The full 162 historical rolling-M2 records were subsequently reproduced under the unchanged evaluator. Baseline tolerances are voltage 1e-8 pu, loading 1e-5 percentage point, normalized J 1e-6, slack P and maximum Q excess 1e-4 MW/Mvar; convergence and stored violation counts match exactly. The native PV/slack Q bounds and controls are preserved. Complete historical generator-Q vectors do not exist in the comparison CSVs, so the historical Q-summary match is not presented as a full device-vector comparison.

Injection algebra and the mapping change

For sigma_n=+1 import/-1 export, A[:,n]=sigma_n*(C_P[:,n]-pi). Participant net P is -C_P diag(sigma)(v-x), with counter-injection pi sum sigma(v-x). Participant Q is C_Q[-sigma*(v-x)*tan(acos(0.98))] for lagging, reversed for leading and zero for unity. The new C_Q equals C_P exactly, all columns are nonnegative and normalized, active injections remain unchanged, and summing columns proves conserved total Q. The spatial difference is checked through a separate participant-by-participant summation. Unity input Q is identical, and independently solved unity outcomes agree numerically.

The solver is pandapower 3.2.1 Newton-Raphson, dc initialization then flat only if necessary, voltage angles calculated, Q limits enforced, connectivity checked, 100 iterations and tolerance_mva=1e-8. Settings and source network parameters are unchanged. Original active devices are replaced only as specified by the historical reconstruction: source PVs and one slack retained, synthetic fixed PQ represents remaining net injections. Slack P covers nonlinear losses. No redispatch optimization is introduced.

Metrics and thermal convention

Line loading is max-end current relative to native current capacity (including derating/parallel factors); transformer loading is the historical current-based runpp convention. The inherited apparent-normalized column name is retained for archive compatibility, but J_AC is the sum of positive excesses of native loading/100 over one. It is dimensionless and differs from the DC sum of MW exceedances against central active-power ratings. Native maximum current exceedances are retained in kA, with explicit terminal limits for transformers.

Bus voltages, counts/deviations outside 0.95-1.05 pu and complete gen/ext-grid Q values versus model Qmin/Qmax are retained. Simple bounds are not complete Q capability curves. Nonconvergence is recorded with attempts and failure cause; a failed solve receives no zero overload score. Thermal, voltage and Q checks are distinct.

Aggregated outcomes

Map	Policy	N	Converged	Thermal_pass	Voltage_pass	Q_pass	Peak_loading	J_AC_sum
Common v2	C	162	162	78	162	162	217.253	50.573
Common v2	EC	162	162	66	162	162	227.377	58.480
Common v2	ROLLING_M2	162	162	84	162	162	212.695	47.442
Common v2	U	162	162	66	162	162	278.395	54.961
Historical	C	162	162	78	162	162	217.171	50.556
Historical	EC	162	162	66	162	162	227.286	58.453
Historical	ROLLING_M2	162	162	84	162	162	212.634	47.430
Historical	U	162	162	66	162	162	278.279	54.933

Paired policy comparisons

Map	A	B	A lower J	B lower J	J ties	Both feasible	Only A	Only B	Neither
Common v2	C	ROLLING_M2	0	15	147	78	0	6	78
Common v2	C	U	18	9	135	66	12	0	84
Common v2	EC	U	18	90	54	54	12	12	84
Historical	C	ROLLING_M2	0	15	147	78	0	6	78
Historical	C	U	18	9	135	66	12	0	84
Historical	EC	U	18	90	54	54	12	12	84

For C versus U, the common-map AC score favors C in 18 cases and U in 9, with 135 ties. For C versus ROLLING_M2, the common-map AC score favors C in 0 cases and ROLLING_M2 in 15, with 147 ties. For EC versus U, the common-map AC score favors EC in 18 cases and U in 90, with 54 ties. The EC annual DC improvement therefore does not establish an improvement in these selected AC checks. These counts concern the fixed stress sample only.

Differences are A minus B on identical year, time, state and PF, always within one mapping. All per-pair score, loading, voltage and Q differences, win/loss/tie classifications and both/only/neither feasibility labels are in MATCHED_AC_PAIRWISE_COMPARISONS.csv. Nonconvergence is separate. Numerical deadbands are 1e-4 J, 1e-4 percentage point peak loading, 1e-6 pu voltage deviation and 1e-6 Mvar Q excess.

Sensitivity to participant Q embedding

Policy	N	Common_lower_J	Historical_lower_J	J_ties	Max_abs_J_change	Max_abs_loading_change_pp	Changed_feasibility
C	162	13	21	128	0.002	0.150	0
EC	162	5	24	133	0.002	0.157	0
ROLLING_M2	162	13	15	134	0.002	0.061	0
U	162	13	31	118	0.002	0.151	0

A	B	Pairs	Strict_J_reversals	Changed_J_class	Changed_feasibility_class
C	U	162	0	0	0
C	ROLLING_M2	162	0	0	0
EC	U	162	0	0	0

Policy	Convergence_changes	Thermal_status_changes	Voltage_status_changes	Q_status_changes	Peak_changes_over_1pp
ROLLING_M2	0	0	0	0	0
U	0	0	0	0	0
C	0	0	0	0	0
EC	0	0	0	0	0

Common minus historical differences for all 648 identical policy/hour/state/PF objects are in AC_MAPPING_SENSITIVITY.csv. Unity cases are unaffected apart from numerical noise. Compare policy ranking blocks explicitly to assess reversals or tie changes; do not infer mapping independence from aggregate totals alone. The detailed CSV preserves adverse and favorable changes alike.

QC, interpretation and verdict

All raw bus/branch/reactive-device rows are independently aggregated against the saved summaries, and all paired keys are unique and complete. Native line loading is independently reconstructed from saved current and current rating. Full historical rolling-M2 comparisons pass the original predeclared reconstruction tolerances. Input hashes and source/code identities remain linked to the configuration.

CONDITIONAL GO for selected fixed-injection AC interpretation only. Lower J is not AC thermal feasibility; convergence is not generator deliverability; repeated contingencies and PF settings are not independent trials. No annual AC-security claim, reserve/ramp verification, prospective policy advantage or AC-constrained optimal-value certificate follows. These results qualify the engineering interpretation while leaving the original DC entitlement geometry and annual numerical certificates unchanged.